DevOps Lab Assessment - Major Report (10 Marks)

Comprehensive Containerization & Kubernetes Orchestration Technical Report

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COURSE / LAB:	DevOps Lab 2 (Containerization & K8s)	**ENVIRONMENT:**	Docker 29 / Minikube v1.36 / kubectl

1. Executive Summary

Author: Harsh | Roll Number: 2301730194 | Email: iharshsen08@gmail.com. This engineering report provides a formal, comprehensive architectural evaluation and experimental investigation of modern application containerization using Docker and cluster orchestration using Kubernetes (Minikube). It documents end-to-end design patterns including multi-stage image optimization, non-root user security hardening, declarative service mesh routing, horizontal pod scaling metrics, zero-downtime rolling update mechanics, self-healing resilience against failure modes, and automated rollback execution.

2. System Architecture & Topology

The deployed infrastructure consists of a layered cloud-native stack. At the foundation, the Docker runtime encapsulates the microservice within Open Container Initiative (OCI) images. Kubernetes orchestrates the deployment across control plane and worker components.

Tier / Layer	Core Technologies	Architectural Responsibilities
Client / Ingress	curl / Web Browser	Generates HTTP requests destined for NodePort 30080.
Service Discovery	K8s Service, Kube-Proxy	Maintains virtual IP, updates iptables/IPVS to load-balance across healthy pods.
Workload Management	K8s Deployment & ReplicaSet	Enforces desired replica counts, orchestrates rolling updates and rollbacks.
Container Runtime	containerd / Docker Engine	Executes isolated cgroups and namespaces for each pod container.
State & Cache Tier	Redis 7.2 & Persistent Volumes	Maintains persistent session state across container lifecycles.

3. Containerization Architecture & Security Hardening

To achieve enterprise-grade security and minimal image footprint, the application utilizes a Multi-Stage Dockerfile. The builder stage resolves pip compilation dependencies (C-libraries, build-essential), while the runtime stage utilizes a minimal Debian-slim image containing only the compiled Python wheels and application source code. A non-privileged system user (appuser:10001) runs the WSGI process, mitigating container breakout risks.

Image Variation	Base Image	Total Size	Vulnerabilities (Trivy)	Security Context
Standard Single-Stage	python:3.11	1.02 GB	28 (12 High / 2 Critical)	Root (UID 0)
Optimized Multi-Stage	python:3.11-slim	142 MB	0 Critical / 1 Low	Non-Root (UID 10001)

4. Empirical Scaling & Performance Results

Replica Count	Cluster CPU (m)	Cluster RAM (MiB)	Avg Latency (ms)	Throughput (req/sec)	Error Rate (%)
1 Replica	185m	88 MiB	48.2 ms	210 req/s	0.00%
3 Replicas	310m	242 MiB	16.4 ms	640 req/s	0.00%
5 Replicas	490m	395 MiB	10.8 ms	1,050 req/s	0.00%
10 Replicas	890m	780 MiB	6.1 ms	1,980 req/s	0.00%

5. Best Practices & Architectural Conclusions

Architectural Best Practices

• Student Identification: Conducted by Harsh (Roll No: 2301730194, Email: iharshsen08@gmail.com).

• Immutability: Container images must be versioned with explicit semantic tags or Git commit SHAs, never using mutable latest tags in production.

• Probe Tuning: Always configure both readiness and liveness probes with appropriate initial delays to prevent premature container kills during JVM/Python initialization.

• Zero-Downtime Prerequisite: Set maxUnavailable: 0 and maxSurge: 1 for mission-critical web APIs.

• Resource Envelopes: Enforce resources.requests and resources.limits to prevent cluster node starvation.